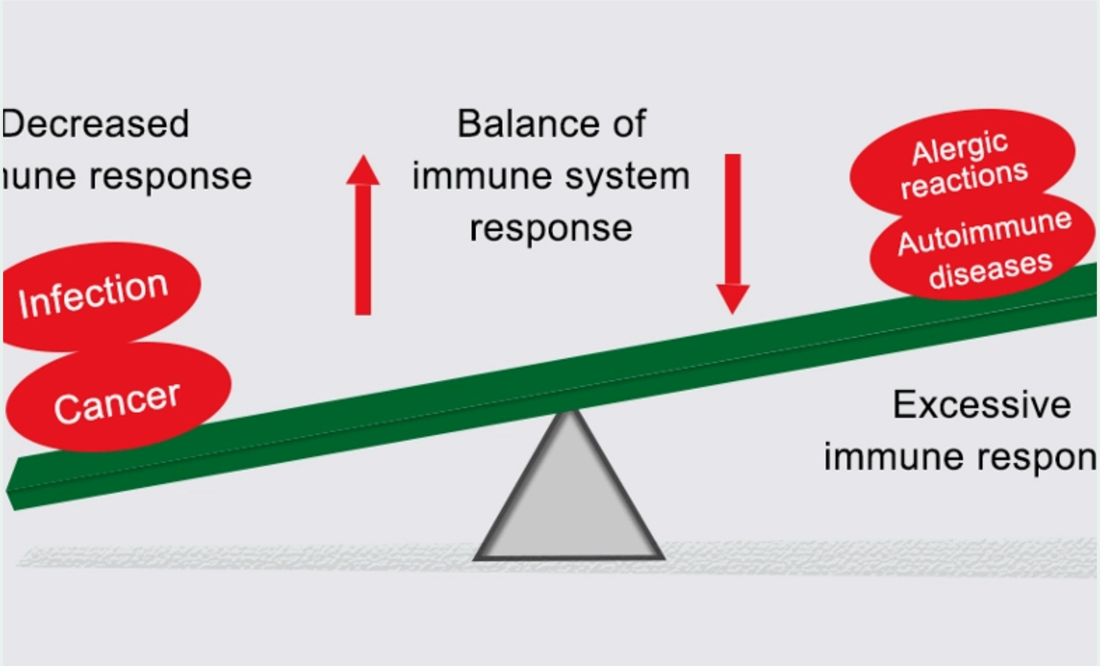
4.7 Immune Tolerance

- Recall two reasons why immune regulation is important.
  - To **avoid excessive lymphocyte activation** and tissue damage
  - To prevent action against self-antigens
- What is autoimmunity?
  - An **immune response against "self"**
- What are the two types of autoimmunity?
  - **Systemic** (i.e Type I diabetes which attacks insulin)
  - **Organ-specific** (i.e multiple sclerosis which attacks general nerve cells)
- Recall three autoimmune diseases.
  - Rheumatoid arthritis (organ-specific)
  - Psoriasis (organ-specific)
  - Lupus (systemic)
  - Grave's disease (systemic)
  - Type I diabetes
  - Multiple sclerosis

▼ Speaking in general terms, what causes autoimmune diseases?

- An **imbalance** between **immune activation** and **control**
- For example, too much activation and too little control can lead to autoimmune diseases



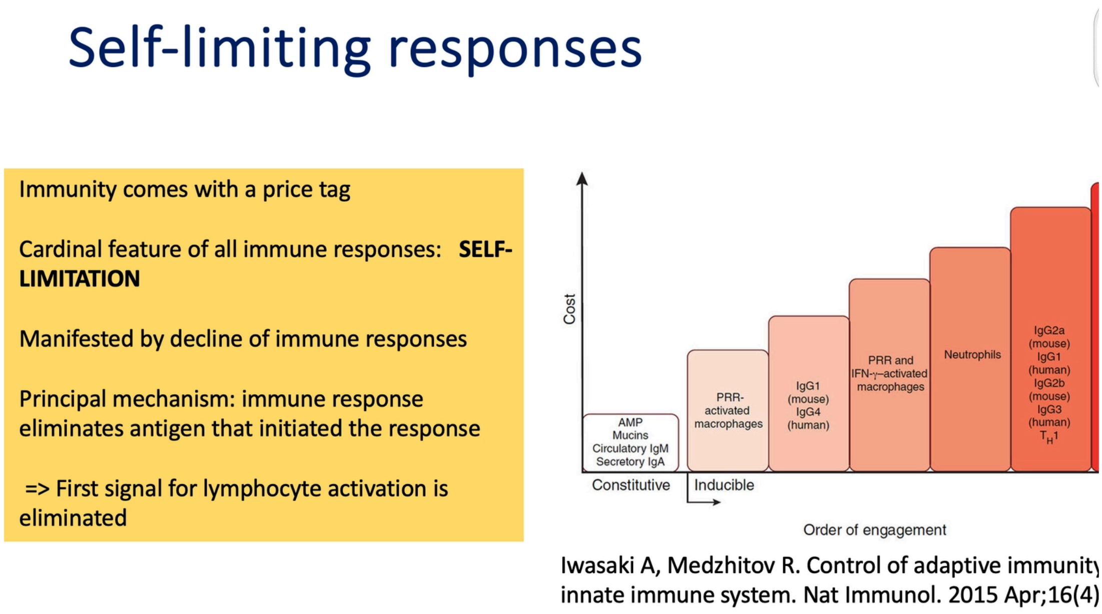
▼ What are the two underlying causative factors of autoimmune diseases?

- Genes that increase susceptibility (i.e certain HLA types can cause the MHC to function abnormally)
- Environmental influences (these influences often combine with the genes to push them to activation)

▼ What problem arises when attacking self antigens that doesn't exist when attacking foreign antigens?

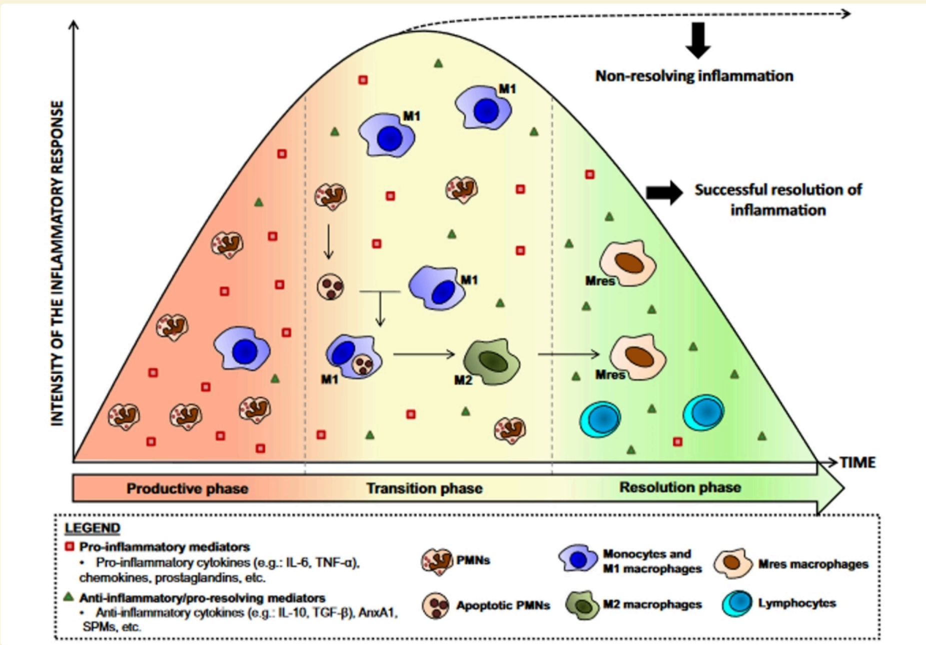
- **Self-antigens are always present**, which means there is a **constant loop of feedback** towards B or T cells which **instructs them to continue attacking or targeting host cells.**
- This means the lymphocytes **never receive the signal** (signal is a lack of antigen) **to stop attacking pathogens** and **to die/convert into memory cells**

- Describe how anaphylaxis arises from a peanut allergy (peanut used as an example).
  - Peanut is consumed by host
  - **Peanut antigen** binds to the **constant region** of IgE antibody (located on mast cell)
  - Mast cell releases **lots of inflammatory mediators** which drives anaphylaxis
- How can T cells be involved in the causation of allergic reactions?
  - T cells are involved in **delayed hypersensitivity** which is when a smaller rash (for example a bee sting) becomes **much bigger over time** due to the **rush and accumulation of T cells at the site**
- How are sepsis and hypercytokinemia linked?
  - **Bacteria from the gut gets into the bloodstream**
  - This leads to a **positive feedback loop of the lymphocytes attacking these bacteria** and it leads to **damage of organs and tissues**
- Diagram explaining how increasingly effective responses against pathogens come at a higher risk of autoimmunity.



- What are the three things that need to occur for a T cell to go from a naive cell to an effector cell?
  - **Recognition** of antigen
  - **Surface contact** with MHC Class I on APC
  - **APCs producing cytokines that activate T cells**

- Extra information on cancer drugs and PD-1.
  - PD-1 is a surface protein on T cells that is used to turn them off and make them inactive
  - Over time, PD-1 increases and active T cell levels decrease
  - Anti-PD-1 drugs are taken for cancer patients which allow all T cells to become active
    - The issue is, since we cannot control the number of T cells which become active, there is a high risk of diseases such as Crohn’s disease and severe skin conditions occurring as the level of active T cells becomes abnormally high.
- What are the three possible outcomes of the immune response to infection?
  - **Resolution** - the pathogen has been successfully dealt with and there was no significant damage to the body
  - **Repair** - The pathogen has caused damage in certain areas of the body and fibroblasts as well as collagen are used to replace the tissues in these areas
    - This can often lead to certain tissues becoming weaker over time, i.e in COPD where the lungs become weaker, leading to breathing difficulties in sufferers
  - **Chronic inflammation -** There is ongoing inflammation at the site of infection and there are attempts to repair the damaged tissue



▼ What is tolerance in immunological terms?

- A **deliberate unresponsiveness** to an antigen that is **induced by the exposure of lymphocytes** to that antigen

▼ What is the difference between central and peripheral tolerance?

- Central tolerance describes tolerance that occurs before B and T cells have been secreted into circulation
- Peripheral tolerance describes tolerance that occurs after B and T cells have been secreted into circulation

▼ What happens in central tolerance?

- The **cells** that can be used to be **destructive against "self"/ human antigens** are **deleted and destroyed before entering into circulation** and causing potential damage.

▼ Describe how central tolerance works in B cell selection.

- Differentiated B cells in the bone marrow go around the bone marrow and patrol for self antigens
- If the bone marrow notices that a B cell receptor recognises a self antigen, then it destroys it before it is released into the bloodstream

▼ Describe how central tolerance works in T cell selection.

- Those **T cells that are unable to bind onto MHCs,** or that **bind to MHCs too strongly**, are **removed from the thymus** and **not released into circulation**
- **AIREs (Autoimmune Regulators)** are transcription factors which are then used to **express the genes of peripheral tissue antigens** in the thymus
- The **expression of these genes** allows us to see **which T cells recognise the self-antigens** and destroy them accordingly.

▼ Why is central tolerance for B cells simpler than for T cells?

- B cells bind to antigens that are free in circulation meaning that whether this happens in the bone marrow or not can be used to select between B cells
- T cells only bind to antigens in circulation on MHC molecules so this makes things more difficult

▼ How can mutations in AIRE affect hosts?

- **AIRE** will hence no longer be able to produce the self-antigens
- It can lead to the **lack of adequate T cell selection** as the **thymus no longer knows which T cells attack self-antigens**
- This results in **multi-organ autoimmunity**

▼ How is peripheral tolerance useful?

- If any T or B cells that attack self antigens **slip through central tolerance by accident**, they can be destroyed via **peripheral tolerance**

▼ What is affinity maturation?

- It is a process where a **B cell receptor can produce antibodies** which **bind onto a pathogen much better** after repeated exposure
- It happens when the **BCR is constantly modified** to **produce slightly different antibodies** and the **lymph nodes select those that are best fit for the antigens**
- It occurs in **germinal centres**

▼ What is a consequence of affinity maturation?

- The **modification of the antibodies** can lead to certain B cells recognising self antigens and attacking self cells.

▼ Describe how anergy works in peripheral tolerance.

- This is when the MHC on the surface of host cells **doesn't contain the necessary stimulatory molecules to activate naive T cells.**
- This means that these self-attacking T cells aren't activated and are less likely to be activated in future.

▼ Describe how ignorance works in peripheral tolerance.

- This is where the **concentration of a self antigen is too low** for T cells to be destroyed (i.e they are ignored)
- This happens in **immunologically privileged areas** (i.e CNS, eye, brain) where the **risk of autoimmunity is greater than the risk of actual infection**

▼ Describe how AICD (activation induced cell death) works in peripheral tolerance.

- Dendritic cells work to turn T cells off instead of turning them on via **apoptosis**

▼ What are regulatory T cells and what do they do?

- They are a **subset of CD4 T cells** which are used to stop the immune response when necessary
- They do this by **releasing the cytokine Interleukin-10 (IL-10)** which can **inactivate lymphocytes.**

▼ What is FoxP3?

- It is a **transcription factor** expressed **to develop CD4 regulatory T cells**
- An **absence of FoxP3** leads to a **reduction in the amount of CD4 regulatory T cells**

▼ Recall two functions of Interleukin-10.

- It is an **anti-inflammatory cytokine**
- It **blocks the synthesis of pro-inflammatory cytokines** (i.e IL-6, IL-8, TNF)
- It **down-regulates macrophage functions.**

▼ What are the two types of T regulatory cells?

- **"Natural" regulatory T cells (nTreg)**
    - Develop in thymus
- **Inducible regulatory T cells (iTreg)**
    - Develop in periphery

▼ Describe how cytokines work.

- They are the chemicals which effectively **dictate which cells are made** in a particular immune response
- They can be **inflammatory or anti-inflammatory**
- They can **control the immune response** to **react in the appropriate manner**

▼ Describe how chemokines work.

- Chemokines work by **driving movement of cells to different areas in the body (chemotaxis)**
- This is done as different chemokines have different receptors on their surfaces.
- For example, a naive T cell can **go to a lymph node via action of a chemokine**, see the antigen it is training to be effective against and **go back to the site of infection via a chemokine**

▼ What component of a T cell allows it to produce cytokines?

- Transcription factors in the nucleus.

▼ What is a T follicular helper cell (TfH)?

- This is a specific type of CD4 T helper cell that plays a critical role by **helping B cells produce antibody against foreign pathogens**
- They communicate with B cells using their cytokines to help them do this

▼ How are T cells directly involved in the production of antibodies?

- When T cells differentiate, they produce cytokines, and different cytokines are produced by different T cells
- These cytokines then tell the B cells to differentiate into B effector cells but specifically they instruct them on which antibody (Ig class) to make
  - For example, if there is a viral infection, Th1 helper cells are produced which produce cytokines that result in IgG production, and this is the antibody that is most effective against viruses.

▼ Diagram of a mast cell.